

1. Combined Nodes Table

Structure

The combined nodes table integrates all nodes (stringers, ribs, spars) into a unified format with special tags to handle specific cases like spars and root nodes.

Column	Description	Notes
1	local_id	Unique identifier for each node.
2	x	X-coordinate of the node.
3	y	Y-coordinate of the node.
4	rib_index	Rib index for the node.
5	stringer_index	Stringer index for the node.
6	tag	Node tag (e.g., "front spars", "rear spars", "stringer").

Special Index Values

- **rib_index:**
 - -1: Root ribs.
 - -2: Front spar ribs.
- **stringer_index:**
 - -1: Front spar stringer.
 - -2: Rear spar stringer.

Purpose

- Centralizes all node data in a unified format for easy access.
- Tags ("front spars", "rear spars", etc.) enable filtering for specific applications.

Example

local_id	x	y	rib_index	stringer_index	tag
1	0.0	0.0	1	1	"stringer"
2	0.1	0.1	2	1	"stringer"
3	0.0	0.0	-1	-1	"front spar"

2. Quadrilateral Surfaces Table

Structure

Each row represents a quadrilateral surface, with detailed metadata.

Column Description Notes

1	local_id	Unique identifier for each surface.
2	node_1	Node ID for the first corner of the quadrilateral.
3	node_2	Node ID for the second corner of the quadrilateral.
4	node_3	Node ID for the third corner of the quadrilateral.
5	node_4	Node ID for the fourth corner of the quadrilateral.
6	stringer_1	Stringer index corresponding to node_1.
7	stringer_2	Stringer index corresponding to node_2.
8	rib_1	Rib index corresponding to node_1.
9	rib_2	Rib index corresponding to node_2.
10	tags	Surface tag (e.g., "quad regular", "quad irregular").
11	area	Calculated area of the quadrilateral.
12	aspect_ratio	Aspect ratio of the quadrilateral (max/min diagonal).

Purpose

- Provides comprehensive metadata for quadrilateral surfaces.
- Facilitates filtering by tags (e.g., "quad regular", "quad irregular").
- Supports validation with geometric metrics (area, aspect_ratio).

Example

local_id	node_1	node_2	node_3	node_4	stringer_1	stringer_2	rib_1	rib_2	tags	area	aspect_ratio
1	101	102	201	202	1	2	1	2	"quad regular"	0.5	1.2

3. Triangular Surfaces Table

Structure

Each row represents a triangular surface, with detailed metadata.

	Column	Description	Notes
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1	local_id	Unique identifier for each triangle.
2	node_1	Node ID for the first vertex of the triangle.
3	node_2	Node ID for the second vertex of the triangle.
4	node_3	Node ID for the third vertex of the triangle.
5	P1 stringer	Stringer index corresponding to node_1.
6	P1 ribs	Rib index corresponding to node_1.
7	P2 stringer	Stringer index corresponding to node_2.
8	P2 ribs	Rib index corresponding to node_2.
9	P3 stringer	Stringer index corresponding to node_3.
10	P3 ribs	Rib index corresponding to node_3.
11	tags	Surface tag (e.g., "tri root", "tri irregular").
12	area	Calculated area of the triangle.
13	aspect_ratio	Aspect ratio of the triangle (longest edge / shortest edge).

Purpose

- Extends metadata to triangular surfaces for root areas and irregular zones.
- Facilitates filtering by tags ("tri root", "tri irregular").
- Supports validation with geometric metrics (area, aspect_ratio).

Example

local_id	node_1	node_2	node_3	P1_stringer	P1_rib_s	P2_stringer	P2_rib_s	P3_stringer	P3_rib_s	tag_s	area	aspect_ratio
1	101	102	201	1	1	2	2	-1	-2	"tri root"	0.3	1.5

Implementation Notes

- Combined Nodes Updates:**
 - Ensure special indices (-1, -2) are included and appropriately tagged.
- Quadrilateral and Triangular Surfaces:**
 - Use functions to dynamically populate metadata and calculate area and aspect_ratio.
- Tags:**
 - Tags like "quad regular", "quad irregular", "tri root", and "tri irregular" should be assigned based on surface type and location.
- Validation:**
 - Use area and aspect_ratio for geometric quality checks.

Future Enhancements

- Connectivity Metadata:**
 - Add columns for adjacent surfaces (e.g., adjacent_surface_1, adjacent_surface_2) for mesh continuity.
- Material and Element Properties:**
 - Include columns like material_id and element_type for FEA-ready surfaces.
- Boundary Conditions:**
 - Add metadata for boundary edges (e.g., bc_edge_1, bc_edge_2) for FEA constraints.